Does a Bag of Tricks Still Work Twenty Years Later? Replicating fastText on AG News, 20 Newsgroups and 2026 News Headlines

Subbareddy Pullalacheruvu
48641626
Macquarie University
subbareddy.pullalacheruvu@students.mq.edu.au

Abstract

I replicate the AG News result of Joulin et al. (2017), who introduced fastText, a linear text classifier that the authors report is about as accurate as much slower deep models. With the authors' own code and hyperparameters, five seeds give 92.34% $\pm$ 0.10 against the paper's 92.5% (with bigrams), and 91.43% against 91.5% (unigrams only). The result replicates. I then test the paper's speed-versus-accuracy claim on data it never used. On 20 Newsgroups (20 classes, long posts, small training set) the paper's configuration reaches only 16.41%. Even after tuning, fastText stays 10.0 points behind a TF-IDF logistic regression baseline, although it is still much faster. On a new dataset of 548 news headlines I collected from RSS feeds in October 2026, a model trained on 2004-era AG News falls from 92.34% to 74.41%. An LLM-as-judge analysis of these errors and a human check of the judge suggest that part of this drop comes from label noise and ambiguous headlines rather than model failure, although the judge and I agreed only moderately on which errors were which. The code, the new dataset and all results are public.[1]

1 Introduction

Open code and data are meant to make results checkable, but a result is only useful if it holds when someone else runs the code and when the data changes. This project asks both questions of one widely used paper. Phase 1 reruns the authors' code on their own AG News data. Phase 2 runs the same pipeline on an existing dataset the paper did not test (20 Newsgroups) and on a dataset I built myself: news headlines from 2026, in the same four classes and format as AG News, collected about twenty years after it. The question running through both phases is whether fastText's "nearly as accurate, much faster" trade-off is a property of the method or a property of AG News.

2 The Paper and Why It Was Chosen

The paper. Joulin et al. (2017) propose a deliberately simple text classifier. Each document is treated as an unordered bag of words and word bigrams. Each word or bigram has a small learned embedding, and bigrams are hashed into a fixed number of buckets so memory stays bounded. The document vector is the average of these embeddings, and a single linear layer with a softmax (or a hierarchical softmax when there are many classes) predicts the label. There are no convolutions, recurrence or attention. The paper claims that on eight standard sentiment and topic datasets this model is "often on par" with deep models while training in seconds on a CPU.[2]

Evaluation framework. The paper reports test *accuracy*, the fraction of test documents given the correct label, on the eight datasets of Zhang et al. (2015), alongside training time. All eight are single-label, so accuracy is well defined, and fastText's precision@1 and recall@1 equal accuracy. On AG News the paper reports 91.5% (unigrams, 10 hidden units, 5 epochs) and 92.5% with bigrams. For comparison, the character-level CNN of Zhang et al. (2015) reaches 87.2% and their word n-gram TF-IDF model 92.4%.

Justification. The paper appeared at EACL 2017, which is CORE-ranked A (ICORE, 2026) and is one of the three general NLP conferences alongside ACL and EMNLP. It has about 5,300 citations on Google Scholar (2026), and the official repository[3] has about 26.5k stars. The code and data are freely available, and the experiments run

[1] https://github.com/subbareddypullalacheruvu/fasttext-replication

[2] https://aclanthology.org/E17-2068/

[3] https://github.com/facebookresearch/fastText

```
"3","Wall St. Bears Claw Back Into
 the Black (Reuters)","Reuters -
 Short-sellers, Wall Street's
 dwindling\band of ultra-cynics,
 are seeing green again."
```

After the authors' `normalize_text`:

```
__label__3 , wall st . bears claw
 back into the black ( reuters ) ,
 reuters - short-sellers , ...
```

Figure 1: An AG News training item as distributed (top) and in fastText input format (bottom). Class 3 is Business.

on a laptop CPU, so a full replication is feasible. One risk was that the repository was archived in March 2024. Its last release (v0.9.2) still installs as `fasttext-wheel` on Python 3.11.

3 The Original Dataset: AG News

AG News (Zhang et al., 2015) was built from a news corpus gathered by the ComeToMyHead academic search engine from more than 2,000 sources, mostly in 2004–2005. It has four balanced classes (World, Sports, Business, Sci/Tech), with 30,000 training and 1,900 test items per class (120,000 / 7,600 in total). Each item is a headline and a short description, usually the first sentence of the article, about 45 tokens on average. The data is distributed as two CSV files with columns *class index, title, description* (Figure 1).[4] I checked the copy I used rather than assuming it was correct. The row counts and per-class counts are exact, there are no empty fields or duplicate rows, and the first record matches the canonical one quoted in other sources.

4 Replication of the Original Work

Environment and procedure. I used the authors' release v0.9.2 through the `fasttext-wheel` Python package, with the hyperparameters from the authors' `classification-results.sh`: `dim=10`, `lr=0.25`, `wordNgrams=2`, `minCount=1`, `bucket=10M`, `epoch=5`, softmax loss. The authors preprocess with a `sed/tr` pipeline (`normalize_text`). I ported it to Python and checked it line by line against the shell output, so the model sees the same tokens. The first setup on Windows 11 failed twice. `fasttext-wheel` needs the Microsoft C++ build tools, and its binding breaks with NumPy 2, so `numpy<2` is pinned. Training is stochastic (Hogwild SGD with random initialisation), so I ran five seeds (0–4) with two threads and report mean ± standard deviation. As a comparison model that the paper does not include, I trained a TF-IDF (word 1–2 grams) logistic regression on the same input.

[4] https://github.com/mhjabreel/CharCnn_Keras/tree/master/data/ag_news_csv

Model	Paper	Mine (5 seeds)	Train
fastText, unigrams	91.5	91.43 ± 0.05	3.2 s
fastText, + bigrams	92.5	92.34 ± 0.10	4.8 s
TF-IDF + LogReg	–	92.58	71.9 s

Table 1: AG News test accuracy (%). "Paper" is Table 1 of Joulin et al. (2017). Training time is wall-clock seconds on a laptop-class CPU.

My first runs (Windows, 9 October 2026) gave three-seed means of 92.25–92.37%. These are archived in the repository. The numbers in this report were regenerated in a single batch with the same package versions (Python 3.11, `fasttext-wheel` 0.9.2, NumPy 1.26) on a Linux machine, so that every result comes from the same run of `results.json`. The two environments agree to within 0.1 points.

Results and closeness. Table 1 shows the result. The bigram model reaches 92.34%, a gap of -0.16 points from the paper. The five seeds span 92.22–92.45%, so the remaining gap is about the size of seed-to-seed variation. The unigram model (gap -0.07) and the gain from adding bigrams (+0.9 points, against +1.0 in the paper) also replicate. The small remaining difference is most likely due to random initialisation and thread scheduling. The paper reports a single run and does not give its seed or thread count. The *efficiency* half of the claim also holds. The TF-IDF model is no more accurate (92.58%) but takes about 15 times longer to train.

Where it works and where it fails. Recall is highest for Sports (98.0%) and lowest for Business (89.3%) and Sci/Tech (89.8%). Most errors are Business ↔ Sci/Tech confusions about technology companies, where the headline mentions a technology firm but the story is about money. Some "errors" are problems in AG News itself. For example, a Tata Consultancy share listing is labelled *World*, and a Nortel revenue warning is labelled *Sci/Tech*. The model's predictions (Business) are arguably better than the gold labels in both cases. Many remaining errors are deals between technology firms, where either label can be defended. *"PeopleSoft, IBM strike middleware alliance"* is gold Sci/Tech but predicted Business, and *"I.B.M. sought a China*

partnership, not just a sale" is gold Business but predicted Sci/Tech. The second also shows a tokenizer side effect that matters later: the authors' `normalize_text` splits "I.B.M." into *i . b . m .*, so the abbreviation never appears as one word.

5 Construction of New Data

5.1 20 Newsgroups (existing dataset)

20 Newsgroups (Lang, 1995) is a collection of Usenet posts from 20 discussion groups. I used the standard "bydate" training portion (11,314 posts) as a single JSON file.[5] It differs from AG News in three ways. It has 20 classes instead of 4. Its posts average 262 tokens instead of 45. And it has 13 times less training data. Raw posts contain `Newsgroups:` headers, email addresses and signatures that give away the label, so I removed headers, signatures and quoted replies with scikit-learn's standard functions. Posts with fewer than three remaining words were dropped. That leaves 10,959 posts, split 80/20 with stratification (seed 42) into 8,769 training and 2,190 test posts.

5.2 UCI News Aggregator (descoped)

My proposal also planned the UCI News Aggregator dataset (Gasparetti, 2017). I wrote its preprocessing, but did not run it, for two reasons. Only two of its four classes (business, sci/tech) overlap with AG News, so it cannot test transfer from AG News the way the 2026 data can. And with the time left, the LLM-as-judge analysis was a better use of it. The code path (`preprocess.py uci`) is in the repository.

5.3 2026 RSS headlines (new dataset)

The point of this dataset is a question AG News cannot answer by itself. AG News describes news language of 2004. Does a model trained on it still work on news written in 2026, where Sci/Tech is dominated by AI and data centres?

Collection. `collect_rss.py` reads 15 public RSS section feeds from the BBC, The Guardian, The New York Times and ESPN. Each item gets the AG News label of the section it was published in. For example, BBC *world* maps to World, Guardian *technology* to Sci/Tech, and NYT *business* to Business. The NYT sports feed returned nothing, so ESPN replaced it. For every item I keep the source, section, link, publication time and collection time.

[5] https://github.com/selva86/datasets

World (BBC *world*) • Suspect linked to Monaco bomb attack on millionaire breaks silence to BBC | Vitalii Zhykovych gives his own account of the attempted assassination…

Sports (BBC *sport*) • Toone comes into England squad as Bronze withdraws | Manchester United midfielder Ella Toone receives a late England call-up…

Sci/Tech (Guardian *technology*) • How Zach Cregger's Resident Evil stands out from all other game adaptations | There's no Leon, Claire or Ada Wong…

Figure 2: Items from the 2026 RSS dataset: label (feed), headline | lead sentence. The third shows a weakness of section labels: a film review filed under technology.

Two collection runs on 9 October 2026 produced 548 unique items.

Processing. I strip HTML and feed boilerplate, and cut the description to its first sentence to match AG News. Duplicates are removed by link and by normalised headline. Then the same `normalize_text` as for AG News is applied. Forty stories appeared in two different sections, for example both World and Business. Their label is ambiguous by construction, so I flag and exclude them. That leaves **508 usable items**: World 127, Sports 135, Business 106, Sci/Tech 140, averaging 37 words (Figure 2).

Annotation and agreement. There was no budget for crowdsourcing, so the labels come from the outlets' sections and are checked by hand. I wrote a one-page labelling guideline that defines each class, gives exclusions and adds a tie-break rule ("what the story is *mainly* about"). I then checked a random sample of 50 usable items against it. 45 of the 50 labels were correct (**90% agreement**), above the 85% threshold I had set in advance for keeping the feed-to-label mapping. The five disagreements were two technology-policy stories filed as Sci/Tech that the guideline puts in World (for example, Finland halting two Google data centres), one arts story that fits no class, one feed item with no real headline, and one environment report. This is single-annotator agreement with the automatic labels, not inter-annotator agreement, because no second annotator was available.

Known limitations. The dataset is small and comes from a single day. It is Anglophone (UK and US outlets). Source and class are confounded: ESPN contributes only Sports and NYT no Sports. Guardian descriptions often run past one sentence.

Data	Model	Acc. (%)	Train
AG News	fastText (paper cfg)	92.34	4.8 s
	TF-IDF + LogReg	92.58	71.9 s
20NG	fastText (paper cfg)	16.41 ± 0.17	2.3 s
	fastText, 25 ep.	62.17 ± 0.18	7.0 s
	+ hier. softmax	56.97 ± 0.42	6.7 s
	fastText, best (100 ep.)	66.50 ± 0.39	20.4 s
	TF-IDF + LogReg	76.53	54.7 s
2026	AG News → 2026	74.41 ± 0.50	–

Table 2: Results on new data, five seeds for fastText. "Paper cfg" is the paper's AG News configuration unchanged. AG News → 2026 is trained on all of AG News and tested on the 508 usable 2026 items.

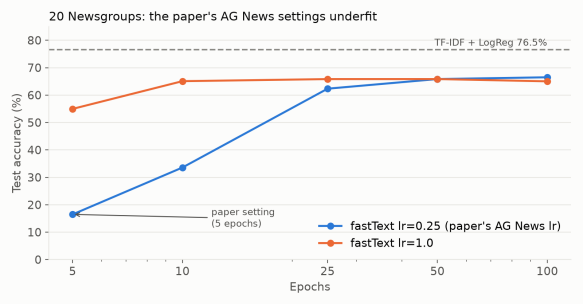


Figure 3: 20 Newsgroups accuracy against training epochs (log scale), five seeds per point. The paper's setting (5 epochs) badly underfits, and even the best setting stays below the TF-IDF baseline.

I froze the dataset at this version for the report rather than keep collecting.

6 Results on New Data

6.1 20 Newsgroups: the configuration does not transfer

With the paper's AG News settings, fastText reaches only 16.41% on 20 Newsgroups (Table 2). It has not failed in principle. It is under-trained: 5 epochs over 8,769 documents is about 14 times fewer document updates than on AG News, and the learning rate decays to zero over those few updates. An epoch and learning-rate sweep (Figure 3) shows accuracy climbing to 66.50% at 100 epochs, lr = 0.25. A higher learning rate (1.0) gets most of the way much sooner: 65.1% after only 10 epochs and 2.8 s. Either way, accuracy then plateaus 10.0 points below TF-IDF logistic regression (76.53%). The *speed* half of the claim survives: fastText still trains several times faster (7.0 s against 54.7 s at 25 epochs). The *accuracy* half does not. On long documents averaging over many words dilutes the few informative ones, and the 10-dimensional hidden layer is too small for 20 overlapping classes. Hierarchical softmax costs accuracy (56.97% against 62.17%) and saves almost no time. The paper uses it for tag prediction with hundreds of thousands of labels, not for 20 classes.

6.2 2026 headlines: a twenty-year shift

A model trained on AG News with the paper's settings reaches 74.41% ± 0.50 on the 2026 headlines, a drop of 17.9 points from its in-domain 92.34%. Recall falls in every class: World 81.7%, Sports 82.1%, Business 59.1%, Sci/Tech 72.0%. Business is hit hardest, with recall falling from 89.3% to 59.1%. Figure 4 shows where accuracy is lost. The 2004 confusions (Business ↔ Sci/Tech) are still there, and new leakage into World appears, because many 2026 Business and Sci/Tech stories are about governments: EU membership, tax pledges, oil prices during a war, and AI in elections. Vocabulary has shifted too. The NYT writes "A.I.", which the authors' tokenizer splits into *a . i .*, so the model never sees the token *ai*. Words such as *openai* (13 items), *chatgpt* and *deepfake* never occur in the AG News training data, so they are out of vocabulary.

Training fastText only on the 2026 data is much worse: 37.22% ± 7.72 from 407 training items, tested on the remaining 101. With the paper's settings and so few items, fastText barely learns. TF-IDF logistic regression trained on the same 407 items reaches 72.28%, close to the transfer result. For fastText, the old large corpus is far better than a small new one, which matches the paper's framing of fastText as a method for large training sets. A 101-item test set is too small for fine comparisons, so these in-domain numbers are indicative only.

6.3 LLM-as-judge: whose mistake is it?

Accuracy treats every disagreement with the gold label as a model error. But the 2026 labels come from newspaper sections, and the spot-check found them only 90% clean. To separate the causes, I used an LLM as a judge (Zheng et al., 2023) for every one of the 127 errors in one seed-0 run. The judge was Claude (`claude-opus-5-5`), given the labelling guideline, the headline and lead sentence, the gold label and the predicted label, but not the outlet. It answered with one verdict. *MODEL_ERROR* means the gold label is clearly right. *AMBIGUOUS* means both labels are defensible. *LABEL_NOISE* means the gold label is wrong or no class fits. It also gave a one-line reason. The

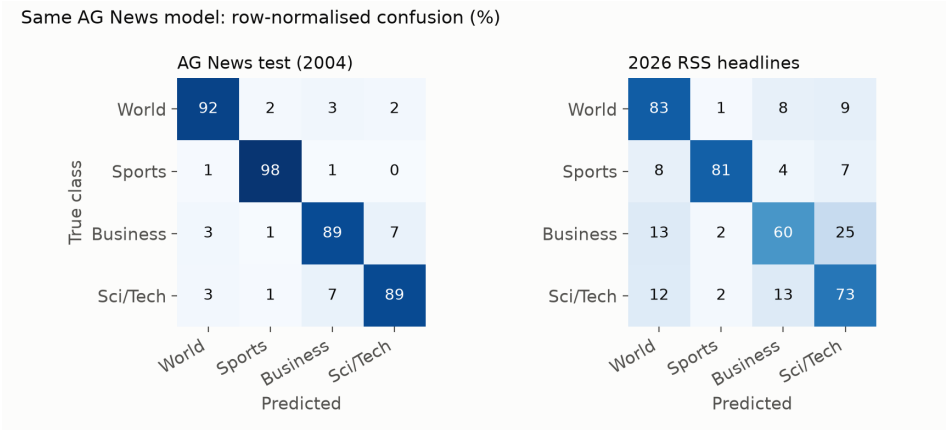


Figure 4: The same AG News-trained model on the AG News test set and on the 2026 headlines. Row-normalised; the diagonal is per-class recall.

prompt and all verdicts are in the repository.

The judge called 55 errors (43%) genuine model errors, 37 (29%) ambiguous and 35 (28%) label noise. For the judged run (75.00%), removing the items the judge called label noise raises accuracy to 80.5%. Also accepting the ambiguous cases raises it to 88.4%. These figures depend on the judge, so they are an estimate rather than a corrected accuracy. The genuine model errors it found concentrate where the vocabulary changed: AI and technology stories (*A.I. agents*, OpenAI's mathematics results), and a video game review that a 2004 model files under World because of the word *war*.

Checking the judge. I labelled a random 30 of the errors myself without seeing the judge's verdicts. We agreed on only 47% of them (Cohen's $\kappa = 0.20$, slight to fair agreement), and on 63% for the simpler question of whether the error is the model's fault. The disagreements were systematic, not random. In 8 of the 16 cases the judge called the gold label correct while I called it label noise. They included three economy stories filed as Business (inflation, oil prices, household income) and several science-section pieces with no clear Sci/Tech topic, such as a hurricane-centre profile. I was stricter than the judge about what the feed sections mean. Both of us still put a large share of the errors outside "clear model error": half of the 30 for the judge, and 20 of the 30 for me. So the conclusion that the raw 17.9-point drop overstates the twenty-year shift holds under either judgement. The exact split between noise, ambiguity and model error does not, because it depends on who judges. The judge is therefore useful for sorting errors at scale, but not as a replacement for a human annotator. Two humans and an adjudicated guideline would be the next step.

6.4 How the new results relate to the replication

On data like AG News (short, balanced, four classes, large training set), the claim holds exactly as published. Change the domain's length, number of classes or training size (20 Newsgroups), and the paper's settings break. Tuning recovers part of the gap, and fastText keeps its speed but not its parity with a stronger linear baseline. Change only the *time* (2026 headlines), and the architecture is unchanged but the learned vocabulary is twenty years out of date. Both experiments evaluate differently from the replication: the 2026 test uses noisier, automatically derived labels, and the 20 Newsgroups setting had to be tuned to be meaningful at all.

7 Ethical Considerations

The original paper. Joulin et al. (2017) has no ethics statement, which was normal before ACL venues introduced ethics review in 2021 (Benotti and Blackburn, 2022). Looking back, using the benefits, harms and vulnerable-groups framing of Benotti and Blackburn: (i) *Benefit*: efficiency is a real ethical gain. Training in seconds on a CPU instead of hours on GPUs reduces energy use (Strubell et al., 2019) and lets researchers without large compute take part. (ii) *Data*: five of its eight datasets (Yelp, Amazon, Yahoo! Answers) are user-generated text whose authors never agreed to become benchmarks and which can contain personal details. AG News is scraped news distributed for non-commercial research. All are mostly English and Western, so "World" means the world as

seen from there. (iii) *Dual use*: classification that is cheap at scale also makes automated content filtering, censorship and surveillance cheap (Weidinger et al., 2021).

Publishing the new data. Under the *Australian Code for the Responsible Conduct of Research* (National Health and Medical Research Council et al., 2018), publishing this work raises the following issues. Following Karamolegkou et al. (2025), I list actions taken, remaining concerns and suggestions, rather than a disclaimer that there are none.

Respect and accountability: copyright. The 2026 dataset contains headlines and lead sentences owned by the BBC, The Guardian, NYT and ESPN, collected from public RSS feeds under each outlet's terms. Copying them for research falls under fair dealing for research or study (*Copyright Act 1968*, s 40). Redistributing them in a public repository is a separate question. I keep the text, with the source link on every row, only to make the results reproducible, and do not claim legal clearance. A version for wider publication should release links, labels and the collection script only, and let users re-collect the text, as is common for social-media datasets.

Respect: privacy. The headlines name real people, including people involved in crime and court stories. The text is already public, but collecting it freezes stories that the publisher may later correct or remove. 20 Newsgroups contains real names and email addresses from the 1990s. Removing headers and signatures was a privacy action as well as a fix for label leakage.

Fairness. All four outlets are UK or US, so the dataset encodes an Anglophone view of what counts as world, business or science news. Arts and culture stories fit none of the four classes and are forced into one or dropped. Source and class are partly confounded, so a model may learn outlet style rather than topic. Adding non-Anglophone outlets and balancing sources per class would reduce this.

Honesty, rigour and transparency. I report the spot-check as single-annotator agreement, not inter-annotator agreement. I report every seed and its spread, not the best run. I keep the earlier Windows results next to the final ones. And I document the LLM judge, its model and its prompt. Using an LLM judge sends dataset text to a third-party service, and its output cannot be exactly reproduced, so the verdicts are stored rather than regenerated. Generative AI (Claude) also helped with code and drafting. I checked all text, numbers and references and am responsible for them.

Human research. No human participants were involved. The only annotation was my own, of public text, so formal human research ethics review was not required.

8 Conclusions and Reflections

The replication worked. With the authors' code and settings, fastText reaches 92.34% on AG News against a published 92.5%, within seed-to-seed variation, and the speed advantage is real. Generalisation is more mixed. The speed claim held everywhere I tested it, but the accuracy claim did not. On 20 Newsgroups the published configuration underfits badly, and even a tuned model trails a TF-IDF baseline by 10.0 points. On 2026 headlines a model trained on 2004 news loses 17.9 points, part of which is label noise in my own data rather than model failure. "About as accurate" turns out to depend on short texts, few classes, plenty of training data and a stable vocabulary.

Three things would change if I started again. First, I would collect the 2026 data over several weeks from the start rather than on one day, since a single news cycle is a thin sample. Second, I would build the labels from a guideline-driven annotation instead of feed sections. Feeds are cheap, but the 40 cross-filed stories and the LLM-judge results show how much of the measured error they cause. Third, I would record every run's seed, thread count and platform from the first day. Rerunning everything in one batch at the end was the only way to get a consistent results file.

Limitations

Replication covers AG News only, not all eight datasets of the original paper. The 2026 dataset is small (508 usable items), comes from one day and four Anglophone outlets, and its labels come from feed sections checked on a 50-item sample by one person. The LLM judge is a single model in a single pass, and it agreed only weakly with my own judgements ($\kappa = 0.20$). The UCI dataset from the proposal was not run.

References

Luciana Benotti and Patrick Blackburn. 2022. Ethics consideration sections in natural language processing

papers. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 4509–4516, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Fabio Gasparetti. 2017. News aggregator data set. UCI Machine Learning Repository.

ICORE. 2026. ICORE conference portal: European chapter of the association for computational linguistics (EACL), ICORE2026 rank A.

Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431, Valencia, Spain. Association for Computational Linguistics.

Antonia Karamolegkou, Sandrine Schiller Hansen, Ariadni Christopoulou, Filippos Stamatiou, Anne Lauscher, and Anders Søgaard. 2025. Ethical concern identification in NLP: A corpus of ACL Anthology ethics statements. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 11618–11635, Albuquerque, New Mexico. Association for Computational Linguistics.

Ken Lang. 1995. NewsWeeder: Learning to filter netnews. In *Proceedings of the Twelfth International Conference on Machine Learning*, pages 331–339.

National Health and Medical Research Council, Australian Research Council, and Universities Australia. 2018. Australian code for the responsible conduct of research.

Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3645–3650, Florence, Italy. Association for Computational Linguistics.

Laura Weidinger, John Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, et al. 2021. Ethical and social risks of harm from language models. *arXiv preprint arXiv:2112.04359*.

Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28*, pages 649–657.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*.